$$i\hbar\gamma^\mu \partial_\mu \psi - mc\psi = 0$$

$$i\hbar\gamma^\mu \partial'_\mu \psi' - mc\psi' = 0 \qquad i\hbar\gamma^\mu \partial_\mu S\psi - mc S\psi = 0$$

$$i\hbar S^{-1}\gamma^\mu \partial_\mu S\psi - mc\psi = 0$$

$$\left(S^{-1}\gamma^\mu \partial'_\mu S - \gamma^\mu \partial_\mu \right) = 0$$

$$S^{-1}\gamma^\mu \partial'_\mu S = \gamma^\mu \partial_\mu$$